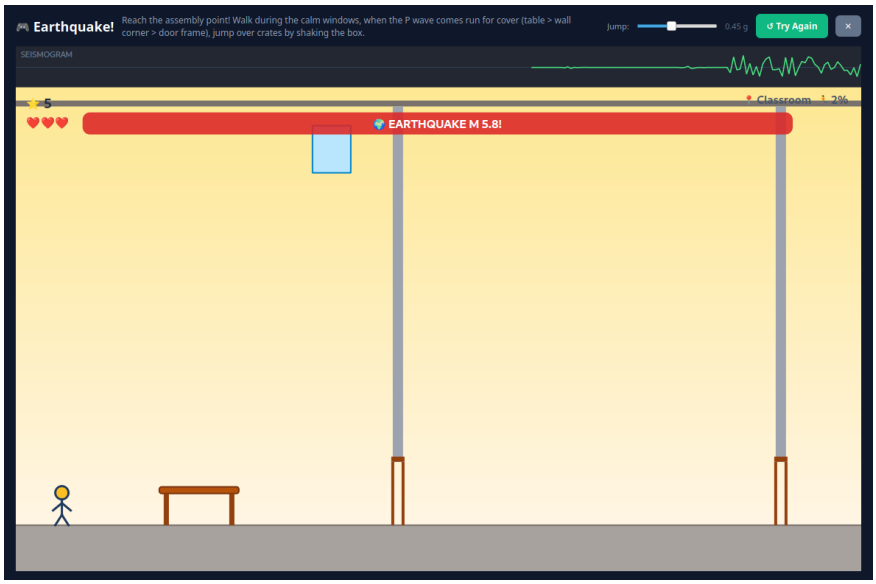
Earthquake!

Crossing the school between aftershocks — and the doorway myth, demolished by the game's own rules

Level: Explorer and Scientist — ages 10–19 • **Time in class:** 20 min as a hook, or a 50-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values shown are simulated; yours will differ.

Before you start

Who it is for

Explorer and Scientist — ages 10–19. The game itself works from about age 10. The **content** connects to earthquakes — geography in **year V** (age 11) and **year IX** — and in the upper school the seismic waves and the early-warning window become quantitative. **It is the only activity in the collection whose outcome is a safety habit** rather than a measurement.

Assumed knowledge

Nothing at all. It helps if the pupils have already done *Seismology* or *The Seismometer*, because they will recognise the seismogram along the top of the screen.

Time in class

Short: 20 min, one round and the discussion about shelters. Full: a 50-minute lesson that ends with **a map of your own school** — where you would take cover in each room you use.

Teacher preparation

10 min, and **two of them are not about the game**: find out **your school's actual evacuation plan** and where **the assembly point** is, so that you can end the lesson with the real information. Then play one round, to see how quickly the aftershocks come.

Sensor settling time

None. The accelerometer responds at once.

Sensors and boxes

MPU-6500. One box is enough, with the screen projected and the children taking turns. The box is **tilted** left and right to walk, and **shaken briefly up and down** to jump.

Three routes

Short	20 min: the prediction about shelters, one projected round, two or three pupils' rounds, and the debrief about the doorway.
Full	The 50-minute lesson: the short route, plus **a map of your own rooms** — the classroom, the hallway, the laboratory — compared with the official evacuation plan.
Project	Extension: why the P wave arrives before the S wave, and how that difference becomes the few seconds of warning.

1. Where this fits the curriculum

- **Geography** — in **year V**, earthquakes: where they happen, why, and **what to do**; in **year IX**, seismicity and seismic risk. The "what to do" part is usually missing from the lesson, and here it is the whole activity.
- **Physics** — in the upper school, wave propagation and the difference between longitudinal and transverse waves: the P wave is faster, arrives first, and the gap before the S wave *is* the warning.
- **Biology and health education** — behaviour in an emergency, and why a practised habit works when thinking freezes.
- **Personal development** — emergency preparedness, the family plan, the meeting point.
- **Across subjects — demolishing a myth through rules rather than contradiction**. Nobody tells the pupils that the doorway is wrong; the game simply gives it the lowest score and the narrowest shelter, and they find out for themselves.

The central skill is **a correct decision made quickly**. By the end of the activity, a pupil should be able to point, from where they are sitting, at the place they would take cover — and say why it is not the door.

2. Timing

Route	Time	What happens
Hook	**20 min**	The prediction about the three shelters (3), one projected round in which you play (4), the pupils' rounds (8), the debrief about the doorway (5).
Full	**50 min**	The short route (20), **a map of the real classroom** (10), the hallway and the laboratory (5), the school's official evacuation plan and assembly point (10), the conclusion (5).

What the game says and what the guidance says: read this before the lesson

The game *lets* you cross the school between aftershocks, and that is a game mechanic, not advice. **The line above the screen used to say "run for cover" — it now says to take cover where you are and stay down**, which is the correct advice; but the mechanic is still a mechanic, and the difference must be stated in class rather than skirted — children remember what they practised.

What is right both in the game and in reality: *during the shaking you do not move.* The game models this well — the character can barely walk during a shock — and the real reason is the same: most injuries happen while trying to move.

The real rule, in three words: drop, cover, hold on — under a sturdy table, one hand on a table leg, and stay there until it stops. If there is no table: get down beside an interior

wall, head protected by your arms.
You move at only two moments: during the few seconds of warning, and then *one or two steps only*, to a shelter right beside you; and **after the shaking has stopped**, towards the exit and the assembly point. **Crossing the school in search of a better shelter is a game element, not advice.**
Say this explicitly at the debrief, in these words: *in the game you run because otherwise it would not be exciting; in real life you take cover where you are.*

3. Materials

What you need	Per group	Notes
MagicBox with MPU-6500	1	Tilted to walk, shaken briefly upwards to jump.
Laptop with the app	1	**WiFi is preferable**: the game asks for broad, repeated movement.
The school's evacuation plan	1	**The most important item on the list.** Ask the school office or the emergency officer for it before the lesson.
Pupil sheet, printed	1 per pupil	The map of the real room is on page two.
Coloured pencils	a few	For the room map: shelters in green, hazards in red.
Neck strap or wrist strap	1	Jumping means shaking the box; a strap prevents a fall.

Preparation. Ten minutes, of which two are the ones that matter: **find out the school's evacuation plan and assembly point**. A lesson about earthquakes that ends without the school's own information is half a lesson. Then play one round, so that you know how quickly the aftershocks come and how little you can move during them.

4. Science background for the teacher

The P wave, the S wave, and where the seconds of warning come from

An earthquake sends out several kinds of wave, and two of them matter here. The **P wave** is longitudinal — it compresses and stretches the rock along its direction of travel — and it is **the fastest**, around 6 km/s in the crust. The **S wave** is transverse, moves the rock at right angles to its travel, goes at about 3.5 km/s and **does almost all the damage**.
Because one is faster than the other, they *separate* along the way: the further you are from the source, the larger the gap between them. And that gap is exactly **the warning window**: if an instrument near the epicentre feels the P wave and sends an electrical signal — which travels far faster than any seismic wave — a distant city can be alerted *before* the S wave arrives.
In Romania this is not theory. The large Vrancea earthquakes occur at **intermediate depth, between about** 60 **and** 200 **km**, about 150 km north-east of Bucharest — the one of 4 March 1977 was M_w 7.4 — and an early-warning system operated by the National Institute for Earth Physics uses precisely this difference. The window for Bucharest is **about** 25 **seconds**. That is not enough to evacuate a school — **it is exactly enough to take cover**, which is also the message of the activity.
In the game, the warning lasts 3 seconds for the first earthquake and 2.5 for the aftershocks, with the estimated magnitude shown on the banner. It is a compression of reality, but with

the right structure: *an announcement first, then the shaking.*

4.1 Why the doorway is a myth — and what the guidance says

"Stand in the doorway" is the advice most parents in Romania know, and it was once reasonable: in old adobe or unreinforced masonry houses, the door frame really was sometimes the part left standing.

In a modern building it is no longer so. The door is no stronger than the wall around it, it protects you from nothing falling from the ceiling, it exposes you to a slamming door, and — most important of all — it leaves you **standing up, in a narrow space, during shaking that throws you off balance**. Current guidance, Romanian and international alike, recommends instead **drop, cover, hold on**: under a sturdy table, holding on to it, until it stops.

The game says the same thing without preaching, which is why it works: the table covers a wide circle and pays double; the corner beside an interior wall covers less and pays one and a half; the door frame covers a *narrow* circle, pays the smallest bonus, and is labelled *weak* on the screen itself. A pupil who chooses the door sees the difference at the first shower of debris.

Do not announce the conclusion at the start. Let them try all three, then ask which worked best — and only then tell them that this is exactly the choice they will make at home if an earthquake comes, and that their grandparents will probably send them towards the door.

What the game does differently from reality — and what was fixed

The aftershocks stay below the main shock. The game still escalates, but towards a ceiling *below* the main shock's magnitude 5.8 — as in reality, where the largest aftershock is on average about a magnitude unit below. **Until recently they escalated to** 7, above the main shock; if anyone in school remembers that from an older lesson, it is worth saying it has changed. The wording for a class is unchanged: *the big earthquake is followed by smaller ones — unpleasant, sometimes dangerous, but smaller.*

The score now measures the evacuation. Most of it is the *distance covered* towards the assembly point, and sheltering well during a shock adds to that, up to a cap. **Standing still earns nothing.** It used to grow with every second spent in the game, so a pupil who hid under a table and never moved beat one who crossed the school — the exact opposite of the lesson. **Judge success by reaching the assembly point** all the same: that is now what the score says too, but the sentence is clearer than a number.

The seismogram records the child's hand too. The trace along the top shows the simulated shaking *plus* the real shaking of the box. If a pupil waggles the box, the line jumps with no earthquake at all — and that is a truth about real seismographs, not a flaw: they record everything, footsteps in the corridor included. It is also the natural bridge to the *Seismology* activity.

5. Running the lesson

1. **The prediction (3 min).** Before playing, the question that holds the whole lesson together: *in an earthquake, where would you take cover — under the table, in the corner beside an interior wall, or in the doorway?* Take it by a show of hands and write it on the sheet. **Do not comment on the vote**, not even with your face.
2. **One projected round (4 min).** Play it yourself. Show deliberately what happens if you stay in the doorway during a strong aftershock, and how little you can move during the shaking.

Technical note — the controls, the three shelters and what happens along the way

The sensor. MPU-6500, the accelerometer. **Walking**: tilt the box left and right; full speed comes at about 25°. **Jumping**: shake the box briefly *upwards*. The jump sensitivity is set by the **Jump** slider in the header, between 0.20 and 0.80 g — **lower it for pupils who cannot shake hard**; it is the most useful setting in the activity.
How to start. From the **Activities** page, on the **Earthquake!** card. You choose a box when it opens.
The route. Five rooms — **Classroom**, **Hallway**, **Laboratory**, **Library**, **Corridor** — and then the yard, with the assembly point. The header shows the current room and the percentage of the journey completed.
The three shelters, and what each is worth. The **sturdy table** — covers the widest circle and pays **double**; the **corner beside an interior wall** — a smaller circle, ×1.5; the **door frame** — **the narrowest circle** and only ×1.2, labelled *weak* on the screen. The difference is in the geometry as well as the score: the table's umbrella is more than twice as wide as the doorway's, so debris that hits you in the doorway misses you under the table.
Two rooms have no shelter at all: the **hallway** and the final **corridor**. When the alarm sounds there, the only answer is to go back to the last door you passed. **This is not a design oversight but the best question of the lesson**: which rooms in our school have no cover?
The windows shed glass during the shocks — it appears as blue fragments. There are four along the route, and they are to be avoided.
The tall bookcase topples at the first shock of magnitude 5 or above, **never gets up again**, and blocks a strip of floor that has to be jumped over. It is exactly what an unsecured cupboard does: first a hazard, then an obstacle on the way out.
The cycle of events. Warning (the P wave, with the estimated magnitude and a countdown), then the shaking, then a calm window of **six to eleven seconds**. The shaking lasts longer the greater the magnitude — about four seconds at 5.8.
During the shaking you can barely move: the speed drops to a quarter of normal in the strong shocks. This is deliberate and it is realistic.
Three lives, with a second and a half of invulnerability after each hit — which is why you sometimes pass through an object and lose nothing. **The game now says why a life was lost**: at the moment of the hit a label rises beside the character naming what struck you — **Crate!**, **Falling glass!**, **Falling debris!** or **Fallen bookcase!** —, and the end screen shows the tally of hits, for example **Hit by:** 2 × falling glass, 1 × rolling crate (or **No hits. Well done!**). For the shelter test this is valuable: the scribe copies it down, and you can see whether a life was lost because of the shelter or for some other reason.
The buttons. Try Again restarts the game from the beginning; × closes the window. There is no export.

3. **The pupils' rounds (8 min).** Taking turns, with the screen projected. The others have roles — see the adaptations box. Ask those watching to **call out the shelter** when the warning appears; it quickly becomes a collective rehearsal.

4. **The debrief about the doorway (5 min).** Go back to the opening vote and compare it with what they found out.

5. **The map of the real room (full route, 10 min).** Each pupil draws on the sheet the room they are sitting in and marks: **in green** where they would take cover, **in red** the windows, the unsecured cupboards and the heavy objects up high. It is the only part of the activity that will still be useful in ten years' time.

6. **The hallway and the laboratory (full route, 5 min).** The same two questions for the rooms with no sturdy tables.

7. **The official plan (full route, 10 min).** The school's assembly point, the route to it, and who gives the signal. **End the lesson with the real information**, not with the score.

Debrief: the score is not the point

Start with the opening vote. *At the start of the lesson, ____ of you chose the doorway. Who would change their mind now, and why?*
This is the only activity in the collection in which a change of opinion *is* the result. Do not rush the moment.
The four questions, in order:

- *Which shelter worked best, and why?* — the table: it covers more and it keeps you down. Let them be the ones to say "it covers more".
- *Why is the doorway the weakest?* — it does not cover you and it leaves you standing. **And why do so many adults still recommend it?** — because it was good advice for old houses. *They were not being foolish; the way we build changed.*
- *What did you do in the hallway, where there was no shelter?* — went back to the last door. **And where is our school's hallway with no cover?**
- *What could move during the shaking — you, or almost nothing?* — almost nothing. Which is why you take cover where you are and leave after it stops.

The correction that must be made — it is the most important sentence of the lesson: *in the game you run through the school because otherwise it would not be a game. In reality, when the shaking starts you take cover where you are, and you move only once it has stopped.*
The close. Each pupil points, from their seat, at the place they would take cover. They are not asked to explain, only to point. A teacher who sees thirty fingers aimed at tables and none at the door has finished the lesson.
And a note on tone. End with something to do, not with fear: *Bucharest is a city with seismic risk, which is why we know what to do, which is why we have a warning system, an evacuation plan and an assembly point.* That is the right ending.

The investigation, in one table

The question. Does the shelter you choose really matter — or is it all luck?
The hypothesis. Written **before** playing: I think the best shelter is _____. It is not changed afterwards.
What I change. One thing only: the shelter chosen. Three rounds, and in each I always take cover at the *same* kind of place — the table only, the corner only, the doorway only.
What I measure. How many lives I have left after three earthquakes, how many points I collected, and the tally of hits on the end screen.
What I keep the same. The same route, the same box, and **going to the shelter as soon as the warning appears**, not at the last moment. If in one round I arrive late, that round does not count.
How many repeats. Three rounds for each kind of shelter — nine in all — if there is time; otherwise divide the shelters between groups and pool the results on the board. **The group version is better**: more data and less boredom.
What makes an observation sound — and what makes it an artefact. Sound is a round in which you reached the shelter before the shaking and stayed there. An artefact is: a round in which a rolling crate hit you, because that has nothing to do with the shelter — the tally at the end now shows what hit you, so it can be checked; a round in which you stayed under the table during the calm as well, because then you are testing nothing; and **any comparison of scores between rounds in which you got different distances along the route**, because the distance covered is most of the score. **Compare lives left** — they depend only on the shelter, which is the variable under test.

6. After the activity

6.1 What the observations should look like

What I do	What happens	What it means
Stay under the table at every shock	Rarely lose lives; double points	The widest cover, and the one recommended in reality.
Stay in the corner beside the wall	Works, but less safely	Less cover; the second choice.
Stay in the doorway	Hit often	**The narrowest shelter** — and the screen itself says "weak".
In the hallway when the alarm sounds	Nowhere to go	Back to the last door. Some rooms have no cover.
Try to cross during the shaking	I can barely move	**The central lesson**: you do not move during an earthquake.
Pass a window during a shock	Glass	Windows are danger zones, in the game and in reality.
First shock of magnitude 5 or more	The bookcase topples	Unanchored furniture becomes a hazard, then an obstacle.
Shake the box for no reason	The seismogram jumps	A seismograph records everything, you included.

6.2 From the game to the school's real plan, step by step

1. **The room you are in now.** Each pupil draws its plan on the sheet, simply: the door, the windows, the desks, the teacher's desk, the cupboards.
2. **Mark in green** the place they would take cover. **Under the desk is a correct answer** — school desks hold.
3. **Mark in red** the windows, the tall unsecured cupboards, the television on the wall, the pots up high and anything else heavy above head height.
4. **Count the red marks.** There are almost always more than the class expected. **This is the moment that stays with them.**
5. **Repeat for the hallway**, where there are no tables — as in the game. What do you do there? Get down beside an interior wall, away from windows and display cases.
6. **Bring out the official evacuation plan** and find the assembly point on it. Compare it with what they drew.
7. **Finish with the route**: how we leave this room, and where we stop. **That is the one sentence of the lesson everybody must know.**

6.3 Questions to ask afterwards

1. Why is a table a better shelter than a doorway? Give two reasons.
2. Your grandparents were probably taught that the doorway is the safe place. Why was that once reasonable advice, and what changed?
3. During the shaking the character can barely move. Why is that a realistic detail, and what follows from it for what you do?

4. The warning system gives Bucharest about 25 seconds. What can be done in that time, and what cannot?
5. Point out, in this room, the place you would take cover and a place you would move away from.

6.4 Assessment ideas as you go

- **Low stakes:** the pupil ranks the three shelters from best to weakest and gives one reason for the first.
- **Standard:** the pupil draws the room map with the shelters and hazards correctly marked and names the school's assembly point.
- **Extension:** the pupil explains where the seconds of warning come from, starting from the difference in speed between the P and S waves, and says what can be done in them.

Common pitfalls and misconceptions

Announcing at the start that the doorway is wrong. It spoils the one real discovery in the lesson. The vote first, the truth at the debrief.
Letting the game pass for an evacuation drill. It is not one. A game in which you run through the school during an earthquake teaches, if left uncorrected, exactly the wrong reflex. **The correction is compulsory**, not optional.
Judging the shelter test by the score. The score includes the distance covered, so two rounds that got different distances do not compare. For comparing shelters, what counts is the lives left.
Turning it into a competition. This is the one activity in the collection where the real stake is not the score, and a league table moves attention exactly where it should not go.
Ending the lesson without the school's real plan. The commonest omission and the most damaging: the pupils leave knowing what to do in a game but not where their own school assembles.
Assuming the game's aftershocks are realistic. They grow from one to the next, to raise the difficulty, although they stay below the main shock. In reality aftershocks become weaker and rarer as time passes.
Shaking the box to make the earthquake look bigger. The seismogram records the child's hand too; it is funny, but say why it happens.
The USB cable. Jumping means shaking the box. Use WiFi.

Safety and classroom management

This is not an evacuation drill and does not replace one. Say so to the class, and say so in the staff room if anyone asks. The official drill is run to the school's procedure, with the emergency service, not in front of a screen.
The subject touches real family experience. In Romania the 1977 earthquake (M_w 7.4) is a living memory in many families, and some pupils will have felt earthquakes themselves. **Announce the subject at the end of the previous lesson**, not on the day, and **allow anyone to opt out without explaining** — offer the scribe's or timekeeper's role to a pupil who would rather not play.
End with what can be done, not with what can happen. The order matters: hazard, then plan, then assembly point. Not the other way round.
Do not ask anyone to recount what their family went through. If a pupil starts of their own accord, listen briefly and bring the discussion back to what we do.
Be ready for the direct question: *is there going to be a big earthquake?* The honest and steadying answer is that *nobody can predict when*, that *this is why we prepare in advance*,

and that *the school has a plan*. Do not promise that it will not happen.
In the room, during the game. Jumping means shaking the box, so leave a metre of clear space around the player and ask them to stay seated with a clear table in front of them.
Do not physically practise taking cover under the desks during this lesson unless the head agrees and the school's procedure provides for it: thirty children diving under desks at once collide with one another. **Pointing is enough**, and it is exactly what the activity asks for.

Adaptations for pupils with special educational needs

The jump slider, lowered. The first adaptation to make and the most effective: in the header, **Jump** can go down to 0.20 g, at which point a very light shake is enough. It is invisible to classmates.
The controls split between two pupils. One tilts the box to walk, the other shakes it to jump, both holding it together. It works well and turns the game into a task of cooperation.
The shelter-caller's role. Whoever is not playing announces aloud, at every warning, which shelter is nearest. It is the most important role in the room, because it turns watching into rehearsing.
Three real roles, rotated. The **player**; the **shelter-caller**; the **scribe**, who records how many lives were left in each round and with which shelter.
The room map, without drawing. For pupils with graphomotor difficulties, the map can be made with stickers on a pre-printed plan, or spoken aloud while pointing — the required outcome is the *decision*, not the drawing.
The screen shakes during the shocks. Seat pupils sensitive to motion further away or give them the scribe's role; and for pupils anxious about the subject, the scribe's role is the right one anyway.

If things do not go to plan

The character does not move. The box is not sending acceleration data: check the connection and the AX, AY, AZ values on the **Live Data** page.
The character will not jump. The **Jump** slider is set too high. Lower it towards 0.20 g. The shake must be *vertical*, short and sharp.
The character runs off to one side by itself. The box is not being held level: straighten it. Walking responds to sideways tilt.
Nobody reaches the assembly point. Remind them of the strategy: **move only during the calm**, and at the warning go straight into the nearest shelter. Anyone trying to cross during the shaking will not get there — which is exactly the point.
One box and thirty pupils. Project the screen, keep the rounds short, and give everyone else the shelter-caller's role — in fact the best version pedagogically, because the whole class makes the decision at every warning.
The box will not connect. Do not postpone this lesson — it is the only one in the collection whose content does not depend on the apparatus. Go straight to the second part: the room map, shelters in green, hazards in red, the school's evacuation plan and the assembly point. The game is the way into the subject; **the map is the lesson**, and it can be done with a pencil.

7. Further resources

Stable, well-established starting points. Search each site for the subject named — these are top-level addresses, not deep links, so they do not rot.

- **General Inspectorate for Emergency Situations (Romania)** — `https://www.igsu.ro` — search for the material on what to do in an earthquake and the family emergency plan. **It is the reference source for the safety part of this lesson.**
- **Fii pregătit** — `https://fiipregatit.ro` — Romania's official preparedness campaign, with material ready to use in class.
- **National Institute for Earth Physics, Romania** — `https://www.infp.ro` — the Vrancea earthquakes, the hazard maps and the early-warning system.
- **USGS Earthquake Hazards** — `https://www.usgs.gov` — search for *drop, cover, hold on* and the explanation of why the doorway is no longer recommended; this is the source that changed the advice internationally.
- **Great ShakeOut** — `https://www.shakeout.org` — the largest school earthquake drill in the world, with classroom material translated into many languages.
- **STEM Learning** — `https://www.stem.org.uk` — search for building earthquake-resistant towers from straws and modelling clay; it goes excellently after this activity and after *Resonance*.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *point out, in this room, the place you would take cover, and say why not the door.* In this activity the exit ticket is the result of the lesson, not a formality.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Here the three strands look like this:

- **Measurement practice** — the pupil tests one shelter per round, enters it at the warning, and records lives left rather than points.
- **Data interpretation** — the pupil ranks the three shelters correctly from their own rounds and explains the order by the cover each gives, not by luck.
- **Scientific reasoning** — the pupil connects the difference in speed between the P and S waves to the seconds of warning, and distinguishes what the game does from what the guidance says.

The rubric itself, with the descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/earthquake.pdf`. It runs to two pages — the opening vote and the table of the three shelters on the first; **the map of the real room** and the school's plan on the second — so it prints double-sided, one sheet per pupil.

Three things to know before printing it. **The opening vote has a row of its own, written before the game**, because the whole activity rests on comparing it with the opinion held at the end — without it, the change is invisible. **The shelter table asks for lives left, not points**, for the reason given above: most of the score is the distance covered, and this test changes the shelter rather than the route. **The lower half of page two is empty, inside a frame**: that is where the pupils draw the room they are sitting in, shelters in green and hazards in red. It is the one page in the whole collection a pupil can put on the fridge at home — and the last line of the sheet, the school's assembly point, is the one worth writing in large letters. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.